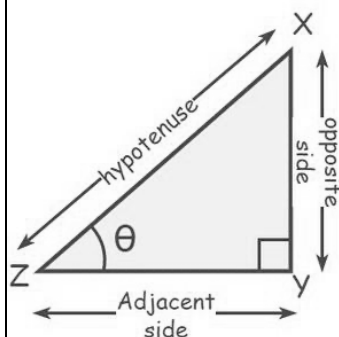


List of Formulae

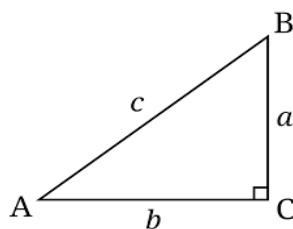
Trigonometry



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$



$$c^2 = a^2 + b^2 \text{ right triangle}$$

Area of triangle

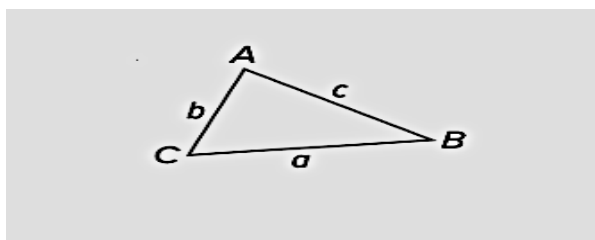
$$A = \frac{1}{2} ab \sin C$$

Law of Sines

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cos C$$



Probability

Independent Events

$$P(B|A) = P(B)$$

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

Conditional Probability Formula

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Addition Principle of Probability

For events A and B in a sample space,
 $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

For mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B)$